

PDF 58, math fragment 1

$$P_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \tag{3.1.1}$$

PDF 58, math fragment 2

$$P_4(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4$$

PDF 58, math fragment 3

$$R_4(x) = e^x - P_4(x) = \frac{1}{120}x^5e^\epsilon, \quad \epsilon \in (-1, 1),$$

PDF 58, math fragment 4

$$|R_4(x)| \leqslant \frac{e}{120}|x^5|, \quad \max_{-1 \leqslant x \leqslant 1} |R_4(x)| \leqslant \frac{e}{120} \approx 0.0226.$$

PDF 58, math fragment 5

$$f(x)$$

PDF 58, math fragment 6

$$f(x)$$

PDF 58, math fragment 7

$$x_0$$

PDF 58, math fragment 8

$$f(x)$$

PDF 58, math fragment 9

$$|x - x_0|$$

PDF 58, math fragment 10

$$f(x) = e^x$$

PDF 58, math fragment 11

$$[-1, 1]$$

PDF 58, math fragment 12

$$e^x$$

PDF 58, math fragment 13

$$e^x$$

PDF 58, math fragment 14

$$A$$

PDF 58, math fragment 15

$$f(x)$$

PDF 58, math fragment 16

$$B$$

PDF 58, math fragment 17

$$P(x) \in B \subseteq A$$

PDF 58, math fragment 18

$$P(x)$$

PDF 58, math fragment 19

$$f(x)$$

PDF 58, math fragment 20

$$A$$

PDF 58, math fragment 21

$$[a, b]$$

PDF 58, math fragment 22

$$C[a, b]$$

PDF 58, math fragment 23

$$B$$

PDF 58, math fragment 24

$$x$$

PDF 58, math fragment 25

$$y$$

PDF 58, math fragment 26

$$-1$$

PDF 58, math fragment 27

$$1$$

PDF 58, math fragment 28

$$O$$

PDF 58, math fragment 29

$$x = -1$$

PDF 58, math fragment 30

$$x = 1$$

PDF 59, math fragment 31

$$\|f(x) - P(x)\|_\infty = \max_{a \leq x \leq b} |f(x) - P(x)|, \tag{3.1.2}$$

PDF 59, math fragment 32

$$\|f(x) - P(x)\|_2 = \sqrt{\int_a^b [f(x) - P(x)]^2 dx}, \tag{3.1.3}$$

PDF 59, math fragment 33

$$\|f(x) - P(x)\|_\infty < \varepsilon$$

PDF 59, math fragment 34

$$\begin{cases} B_n(f, x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) P_k(x), \\ P_k(x) = \binom{n}{k} x^k (1-x)^{n-k}, \end{cases} \tag{3.1.4}$$

PDF 59, math fragment 35

$$\lim_{n \rightarrow \infty} B_n^{(m)}(f, x) = f^{(m)}(x).$$

PDF 59, math fragment 36

$$L_n(x) = \sum_{k=0}^n f(x_k) l_k(x), \quad \sum_{k=0}^n l_k(x_k) = 1$$

PDF 59, math fragment 37

$$\sum_{k=0}^n P_k(x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} = 1. \tag{3.1.5}$$

PDF 59, math fragment 38

$$\|\cdot\|_\infty$$

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PDF 59, math fragment 58

$$\|\cdot\|_2$$

$$P_n(x)$$

$$f(x)\in C[a,b]$$

$$f(x)\in C[a,b]$$

$$f(x)\in C[a,b]$$

$$[a,b]$$

$$P_n(x)$$

$$f(x)$$

$$[a,b]$$

$$f(x)$$

$$P_n(x)$$

$$f(x)$$

$$f(x)\in C[a,b]$$

$$\varepsilon>0$$

$$P(x)$$

$$[a,b]$$

$$\lim_{n\rightarrow\infty}B_n(f,x)=f(x)$$

$$[0,1]$$

$$f(x)$$

$$[0,1]$$

PDF 59, math fragment 59

$$m$$

PDF 59, math fragment 60

$$f(x)$$

PDF 59, math fragment 61

$$B_n(f,x)$$

PDF 59, math fragment 62

$$f(x)=1$$

PDF 59, math fragment 63

$$\sum_{k=0}^nl_k(x_k)=1$$

PDF 59, math fragment 64

$$l_k(x_k)=1$$

PDF 59, math fragment 65

$$n+1$$

PDF 59, math fragment 66

$$\sum_{k=0}^nl_k(x)=1$$

PDF 60, math fragment 67

$$(x+y)^n=\sum_{k=0}^n\binom{n}{k}x^ky^{n-k}\tag{3.1.6}$$

PDF 60, math fragment 68

$$\sum_{k=0}^n|P_k(x)|=\sum_{k=0}^nP_k(x)=1$$

PDF 60, math fragment 69

$$|B_n(f,x)|\leqslant\max_{0\leqslant x\leqslant 1}|f(x)|\sum_{k=0}^n|P_k(x)|\leqslant\delta$$

PDF 60, math fragment 70

$$\|f\|_\infty=\max_{a\leqslant x\leqslant b}|f(x)|,$$

PDF 60, math fragment 71

$$\|f+g\|\leqslant\|f\|+\|g\|\tag{3.1.7}$$

PDF 60, math fragment 72

$$D(f,g)=\|f-g\|_\infty\tag{3.1.8}$$

PDF 60, math fragment 73

$$D(f,g)\leqslant D(f,h)+D(h,g)\tag{3.1.9}$$

PDF 60, math fragment 74

$$\bigl|\|f\|_\infty-\|g\|_\infty\bigr|\leqslant\|f-g\|_\infty\tag{3.1.10}$$

PDF 60, math fragment 75

$$y=1-x$$

PDF 60, math fragment 76

$$x\in[0,1]$$

PDF 60, math fragment 77

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PDF 60, math fragment 91

PDF 60, math fragment 92

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PDF 60, math fragment 95

$$P_k(x) \geqslant 0$$

$$|f(x)| \leqslant \delta$$

$$x \in [0,1]$$

$$B_n(f,x)$$

$$L_n(x)$$

$$\sum_{k=0}^n |l_k(x)|$$

$$B_n(f,x)$$

$$C[a,b]$$

$$[a,b]$$

$$C[a,b]$$

$$f \in C[a,b]$$

$$\|\cdot\|_\infty$$

$$\infty$$

$$\|\cdot\|$$

$$\|f\| \geqslant 0$$

$$f \equiv 0$$

$$\|f\| = 0$$

$$\|af\| = |a|\|f\|$$

$$f \in C[a,b]$$

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PDF 61, math fragment 110

PDF 61, math fragment 111

PDF 61, math fragment 112

PDF 61, math fragment 113

PDF 61, math fragment 114

PDF 61, math fragment 115

$$a$$

$$f,g\in C[a,b]$$

$$C[a,b]$$

$$f\in C[a,b]$$

$$f,g\in C[a,b]$$

$$f$$

$$g$$

$$n$$

$$P_n(x)=a_0+a_1x+\cdots+a_nx^n,$$

$$\max_{a\leqslant x\leqslant b}|f(x)-P_n^*(x)|=\min_{P_n\in H_n}\max_{a\leqslant x\leqslant b}|f(x)-P_n(x)|,$$

$$\Delta(f,P_n)=\|f-P_n\|_\infty=\max_{a\leqslant x\leqslant b}|f(x)-P_n(x)|\tag{3.2.1}$$

$$E_n=\inf_{P_n\in H_n}\{\Delta(f,P_n)\}=\inf_{P_n\in H_n}\max_{a\leqslant x\leqslant b}|f(x)-P_n(x)|\tag{3.2.2}$$

$$P_n^*(x)\in H_n,\Delta(f,P_n^*)=E_n\tag{3.2.3}$$

$$\|f(x)-P_n^*(x)\|_\infty=E_n.$$

$$|P(x_0)-f(x_0)|=\max_{a\leqslant x\leqslant b}|P(x)-f(x)|=\mu,$$

$$|P(x_0)-f(x_0)|=\mu,$$

$$n$$

$$n$$

$$H_n$$

$$H_n\subseteq C[a,b]$$

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PDF 61, math fragment 130

PDF 61, math fragment 131

PDF 61, math fragment 132

PDF 61, math fragment 133

PDF 61, math fragment 134

PDF 61, math fragment 135

$$H_n = \text{span}\{1, x, \cdots, x^n\}$$

$$1, x, \cdots, x^n$$

$$[a, b]$$

$$H_n$$

$$H_n$$

$$P_n(x)$$

$$a_0, a_1, \cdots, a_n$$

$$H_n$$

$$P_n^*(x)$$

$$f(x) \in C[a, b]$$

$$P_n(x) \in H_n, f(x) \in C[a, b]$$

$$f(x)$$

$$P_n(x)$$

$$[a, b]$$

$$\Delta(f, P_n) \geq 0, \Delta(f, P_n)$$

$$\{\Delta(f, P_n)\}$$

$$E_n$$

$$f(x)$$

$$[a, b]$$

$$f(x) \in C[a, b]$$

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PDF 61, math fragment 142

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PDF 62, math fragment 155

$$P_n^*(x)$$

$$f(x)$$

$$[a,b]$$

$$f(x)\in C[a,b]$$

$$P_n^*(x)\in H_n$$

$$f(x)\in C[a,b],P(x)\in H_n$$

$$x=x_0$$

$$x_0$$

$$P(x)$$

$$P(x_0)-f(x_0)=\mu$$

$$x_0$$

$$P(x_0)-f(x_0)=-\mu$$

$$x_0$$

$$P(x)-f(x)$$

$$[a,b]$$

$$x_0\in[a,b]$$

$$P(x)$$

$$P(x)-f(x)>-E_n.$$

$$-E_n+2h\leqslant P(x)-f(x)\leqslant E_n,$$

$$-E_n+h\leqslant [P(x)-h]-f(x)\leqslant E_n-h,$$

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$$|[P(x)-h]-f(x)|\leqslant E_n-h.$$

PDF 62, math fragment 157

$$y=f(x)+E_n\quad\&\quad y=f(x)-E_n$$

PDF 62, math fragment 158

$$\begin{aligned} P(x_k)-f(x_k)&=(-1)^k\sigma\|P(x)-f(x)\|_\infty,\\ \sigma&=\pm1,\quad k=1,2,\cdots,n+2, \end{aligned}\tag{3.2.4}$$

PDF 62, math fragment 159

$$\|f(x)-Q(x)\|_\infty<\|f(x)-P(x)\|_\infty.$$

PDF 62, math fragment 160

$$P(x)\in H_n$$

PDF 62, math fragment 161

$$f(x)\in C[a,b]$$

PDF 62, math fragment 162

$$P(x)$$

PDF 62, math fragment 163

$$P(x)$$

PDF 62, math fragment 164

$$f(x)$$

PDF 62, math fragment 165

$$\mu=E_n$$

PDF 62, math fragment 166

$$P(x)$$

PDF 62, math fragment 167

$$[a,b]$$

PDF 62, math fragment 168

$$x\in[a,b]$$

PDF 62, math fragment 169

$$P(x)-f(x)$$

PDF 62, math fragment 170

$$[a,b]$$

PDF 62, math fragment 171

$$-E_n$$

PDF 62, math fragment 172

$$-E_n+2h$$

PDF 62, math fragment 173

$$h>0$$

PDF 62, math fragment 174

$$x\in[a,b]$$

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$$P(x) - h$$

$$f(x)$$

$$E_n$$

$$E_n$$

$$[a, b]$$

$$y = P(x)$$

$$[a, b]$$

$$P(x)$$

$$y = f(x) - E_n$$

$$y = P(x)$$

$$y = f(x)$$

$$y = P(x) - h$$

$$x$$

$$y$$

$$O$$

$$a$$

$$b$$

$$y = f(x) + E_n$$

$$y = P(x)$$

$$y = f(x) - E_n$$

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$$P(x) \in H_n$$

PDF 62, math fragment 196

$$f(x) \in C[a, b]$$

PDF 62, math fragment 197

$$P(x)$$

PDF 62, math fragment 198

$$[a, b]$$

PDF 62, math fragment 199

$$n + 2$$

PDF 62, math fragment 200

$$n + 2$$

PDF 62, math fragment 201

$$a \leq x_1 < x_2 < \cdots < x_{n+2} \leq b$$

PDF 62, math fragment 202

$$[a, b]$$

PDF 62, math fragment 203

$$n + 2$$

PDF 62, math fragment 204

$$P(x)$$

PDF 62, math fragment 205

$$f(x)$$

PDF 62, math fragment 206

$$[a, b]$$

PDF 62, math fragment 207

$$Q(x) \in H_n, Q(x) \not\equiv P(x)$$

PDF 62, math fragment 208

$$P(x) - Q(x) = [P(x) - f(x)] - [Q(x) - f(x)]$$

PDF 62, math fragment 209

$$x_1, x_2, \cdots, x_{n+2}$$

PDF 62, math fragment 210

$$P(x_k) - f(x_k)$$

PDF 62, math fragment 211

$$k = 1, \cdots, n + 2$$

PDF 62, math fragment 212

$$P(x) -$$

PDF 63, math fragment 213

$$-E_n \leq P(x) - f(x) \leq E_n, \quad -E_n \leq Q(x) - f(x) \leq E_n,$$

PDF 63, math fragment 214

$$-E_n \leq \frac{P(x) + Q(x)}{2} - f(x) \leq E_n.$$

PDF 63, math fragment 215

$$R(x_k)-f(x_k)=(-1)^k\sigma E_n\quad(k=1,2,\cdots,n+2),$$

PDF 63, math fragment 216

$$E_n=|R(x_k)-f(x_k)|=\left|\frac{P(x_k)-f(x_k)}{2}+\frac{Q(x_k)-f(x_k)}{2}\right|. \tag{3.2.5}$$

PDF 63, math fragment 217

$$\frac{P(x_k)-f(x_k)}{2}=\frac{Q(x_k)-f(x_k)}{2}=\pm\frac{E_n}{2}$$

PDF 63, math fragment 218

$$P(x_k)-f(x_k)=Q(x_k)-f(x_k).$$

PDF 63, math fragment 219

$$Q(x)$$

PDF 63, math fragment 220

$$n+2$$

PDF 63, math fragment 221

$$(a,b)$$

PDF 63, math fragment 222

$$n+1$$

PDF 63, math fragment 223

$$P(x)-Q(x)\not\equiv 0$$

PDF 63, math fragment 224

$$n$$

PDF 63, math fragment 225

$$n$$

PDF 63, math fragment 226

$$P(x)$$

PDF 63, math fragment 227

$$P(x)$$

PDF 63, math fragment 228

$$f(x)$$

PDF 63, math fragment 229

$$y=P(x)-f(x)$$

PDF 63, math fragment 230

$$f(x)\in C[a,b]$$

PDF 63, math fragment 231

$$H_n$$

PDF 63, math fragment 232

$$H_n$$

PDF 63, math fragment 233

$$P(x)$$

PDF 63, math fragment 234

$$Q(x)$$

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PDF 63, math fragment 254

$$x\in[a,b]$$

$$R(x)=\frac{Q(x)+P(x)}{2}$$

$$H_n$$

$$R(x)-f(x)$$

$$n+2$$

$$\{x_k\}$$

$$|P(x_k)-f(x_k)|\leqslant E_n,\quad |Q(x_k)-f(x_k)|\leqslant E_n$$

$$P(x_k)=Q(x_k)$$

$$k=1,2,\cdots,n+2$$

$$P(x)-Q(x)$$

$$n+2$$

$$Q(x)\equiv P(x)$$

$$f(x)\in C[a,b]$$

$$P_n^*(x)\in H_n$$

$$f(x)$$

$$P_n^*(x)-f(x)$$

$$[a,b]$$

$$n+2$$

$$n+1$$

$$x_k\leqslant \overline{x_k}\leqslant x_{k+1}$$

PDF 63, math fragment 255

$$k=1,2,\cdots,n+1$$

PDF 63, math fragment 256

$$P_n^*(\overline{x_k})-f(\overline{x_k})=0$$

PDF 63, math fragment 257

$$\overline{x_k}$$

PDF 63, math fragment 258

$$P_n^*(x)$$

PDF 63, math fragment 259

$$P(x)$$

PDF 63, math fragment 260

$$P(x)$$

PDF 63, math fragment 261

$$n=1$$

PDF 63, math fragment 262

$$f(x)\in C^2[a,b]$$

PDF 63, math fragment 263

$$f''(x)$$

PDF 63, math fragment 264

$$(a,b)$$

PDF 64, math fragment 265

$$P_1(a)-f(a)=P_1(b)-f(b)=-[P_1(x_2)-f(x_2)].$$

PDF 64, math fragment 266

$$\begin{cases} a_0+a_1a-f(a)=a_0+a_1b-f(b),\\ a_0+a_1a-f(a)=f(x_2)-(a_0+a_1x_2). \end{cases} \tag{3.2.6}$$

PDF 64, math fragment 267

$$a_1=\frac{f(b)-f(a)}{b-a}=f'(x_2), \tag{3.2.7}$$

PDF 64, math fragment 268

$$a_0=\frac{f(a)+f(x_2)}{2}-\frac{f(b)-f(a)}{b-a}\frac{a+x_2}{2}. \tag{3.2.8}$$

PDF 64, math fragment 269

$$y=\frac{1}{2}[f(a)+f(x_2)]+a_1\left(x-\frac{a+x_2}{2}\right).$$

PDF 64, math fragment 270

$$a_1=\sqrt{2}-1\approx0.414,\quad f'(x)=\frac{x}{\sqrt{1+x^2}},$$

PDF 64, math fragment 271

$$\frac{x_2}{\sqrt{1+x_2^2}}=\sqrt{2}-1,$$

PDF 64, math fragment 272

$$x_2=\sqrt{\frac{\sqrt{2}-1}{2}}\approx0.4551,\quad f(x_2)=\sqrt{1+x_2^2}\approx1.0986.$$

PDF 64, math fragment 273

$$a_0=\frac{1+\sqrt{1+x_2^2}}{2}-a_1\frac{x_2}{2}\approx 0.955,$$

PDF 64, math fragment 274

$$P_1(x)=0.955+0.414x,$$

PDF 64, math fragment 275

$$\sqrt{1+x^2}\approx 0.955+0.414x,\quad 0\leqslant x\leqslant 1. \tag{3.2.9}$$

PDF 64, math fragment 276

$$\max_{0\leqslant x\leqslant 1}|\sqrt{1+x^2}-P_1(x)|\leqslant 0.045.$$

PDF 64, math fragment 277

$$P_1(x)=a_0+a_1x$$

PDF 64, math fragment 278

$$a\leqslant x_1<x_2<x_3\leqslant b$$

PDF 64, math fragment 279

$$P_1(x_k)-f(x_k)=(-1)^k\sigma\max_{a\leqslant x\leqslant b}|P_1(x)-f(x)|$$

PDF 64, math fragment 280

$$\sigma=\pm 1,k=1,2,3$$

PDF 64, math fragment 281

$$f''(x)$$

PDF 64, math fragment 282

$$[a,b]$$

PDF 64, math fragment 283

$$f'(x)$$

PDF 64, math fragment 284

$$f'(x)-a_1$$

PDF 64, math fragment 285

$$(a,b)$$

PDF 64, math fragment 286

$$x_2$$

PDF 64, math fragment 287

$$P_1'(x_2)-f'(x_2)=a_1-f'(x_2)=0$$

PDF 64, math fragment 288

$$f'(x_2)=a_1$$

PDF 64, math fragment 289

$$x_0=a,x_1=b$$

PDF 64, math fragment 290

$$P_1(x)$$

PDF 64, math fragment 291

$$y=P_1(x)$$

PDF 64, math fragment 292

$$MN$$

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MQ

D

x

y

O

a

x_2

b

M

N

Q

D

$y = P_1(x)$

MN

Q

D

MQ

$f(x) = \sqrt{1 + x^2}$

$[0, 1]$

$\sqrt{1 + x^2}$

PDF 64, math fragment 313

$$x_1 < x_2 < x_3$$

PDF 64, math fragment 314

$$x_0 = a, x_1 = b$$

PDF 64, math fragment 315

$$x_1 = a, x_3 = b$$

PDF 64, math fragment 316

$$P_1(x) = 0.955 + 0.414x$$

PDF 64, math fragment 317

$$0.045$$

PDF 64, math fragment 318

$$x = 1$$

PDF 64, math fragment 319

$$\sqrt{2} - 1.369 \approx 0.04521356 > 0.045$$

PDF 65, math fragment 320

$$\sqrt{a^2 + b^2} \approx 0.955a + 0.414b.$$

PDF 65, math fragment 321

$$\|f - P_n^*\|_2 = \sqrt{\int_a^b [f(x) - P_n^*(x)]^2 dx} = \inf_{P \in H_n} \|f - P\|_2, \tag{3.3.1}$$

PDF 65, math fragment 322

$$\int_a^b g(x)\rho(x)dx = 0, \tag{3.3.2}$$

PDF 65, math fragment 323

$$(f, g) = \int_a^b \rho(x)f(x)g(x)dx \tag{3.3.3}$$

PDF 65, math fragment 324

$$\boldsymbol{f} = (f_1, f_2, \cdots, f_n)^{\text{T}}, \quad \boldsymbol{g} = (g_1, g_2, \cdots, g_n)^{\text{T}},$$

PDF 65, math fragment 325

$$x = \frac{b}{a} \leqslant 1$$

PDF 65, math fragment 326

$$f(x) \in C[a, b]$$

PDF 65, math fragment 327

$$P_n^*(x) \in H_n$$

PDF 65, math fragment 328

$$P_n^*(x)$$

PDF 65, math fragment 329

$$f(x)$$

PDF 65, math fragment 330

$$[a, b]$$

PDF 65, math fragment 331

$$P_n^*(x)$$

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$$P_n^*(x)$$

$$(a,b)$$

$$\rho(x)$$

$$\rho(x)$$

$$(a,b)$$

$$1^\circ \int_a^b |x|^n \rho(x) \mathrm{d} x (n=0,1,\cdots)$$

$$2^\circ$$

$$g(x)$$

$$(a,b)$$

$$g(x)\equiv 0$$

$$f(x),g(x)\in C[a,b],\rho(x)$$

$$[a,b]$$

$$f(x)$$

$$g(x)$$

$$[a,b]$$

$$(f,g)=(g,f)$$

$$(cf,g)=c(f,g),c$$

$$(f_1+f_2,g)=(f_1,g)+(f_2,g)$$

$$(f,f)\geqslant 0$$

$$f=0$$

PDF 65, math fragment 352

$$(f, f) = 0$$

PDF 65, math fragment 353

$$C[a, b]$$

PDF 65, math fragment 354

$$n$$

PDF 65, math fragment 355

$$\mathbf{R}^n$$

PDF 66, math fragment 356

$$\|\boldsymbol{f}\|_2 = \left(\sum_{k=1}^n f_k^2 \right)^{\frac{1}{2}},$$

PDF 66, math fragment 357

$$\|f\|_2 = \sqrt{\int_a^b \rho(x) f^2(x) dx} = \sqrt{(f, f)} \tag{3.3.4}$$

PDF 66, math fragment 358

$$|(f, g)| \leq \|f\|_2 \|g\|_2 \quad (\text{Cauchy-Schwarz 不等式}); \tag{3.3.5}$$

PDF 66, math fragment 359

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2 \quad (\text{三角不等式}); \tag{3.3.6}$$

PDF 66, math fragment 360

$$0 \leq (f + \lambda g, f + \lambda g) = (f, f) + 2\lambda(f, g) + \lambda^2(g, g).$$

PDF 66, math fragment 361

$$\|f\|_2^2 - 2\frac{|(f, g)|^2}{\|g\|_2^2} + \frac{|(f, g)|^2}{\|g\|_2^2} \geq 0, \quad |(f, g)|^2 \leq \|f\|_2^2 \|g\|_2^2.$$

PDF 66, math fragment 362

$$\|f + g\|_2^2 = (f + g, f + g) = (f, f) + 2(f, g) + (g, g)$$

PDF 66, math fragment 363

$$\leq \|f\|_2^2 + 2|(f, g)| + \|g\|_2^2$$

PDF 66, math fragment 364

$$\leq \|f\|_2^2 + 2\|f\|_2 \|g\|_2 + \|g\|_2^2$$

PDF 66, math fragment 365

$$= (\|f\|_2 + \|g\|_2)^2,$$

PDF 66, math fragment 366

$$(f, g) = \int_a^b \rho(x) f(x) g(x) dx = 0, \tag{3.3.7}$$

PDF 66, math fragment 367

$$(\varphi_j, \varphi_k) = \int_a^b \rho(x) \varphi_j(x) \varphi_k(x) dx = \begin{cases} 0, & j \neq k, \\ A_k > 0, & j = k, \end{cases} \tag{3.3.8}$$

PDF 66, math fragment 368

$$(\boldsymbol{f}, \boldsymbol{g}) = \sum_{k=1}^n f_k g_k$$

PDF 66, math fragment 369

$$\boldsymbol{f} \in \mathbf{R}^n$$

PDF 66, math fragment 370

$$f(x)\in C[a,b]$$

PDF 66, math fragment 371

$$f(x)$$

PDF 66, math fragment 372

$$f,g\in C[a,b]$$

PDF 66, math fragment 373

$$1^\circ$$

PDF 66, math fragment 374

$$2^\circ$$

PDF 66, math fragment 375

$$3^\circ\|f+g\|_2^2+\|f-g\|_2^2=2(\|f\|_2^2+\|g\|_2^2)$$

PDF 66, math fragment 376

$$g=0$$

PDF 66, math fragment 377

$$g\neq 0$$

PDF 66, math fragment 378

$$\lambda$$

PDF 66, math fragment 379

$$\lambda=-\frac{(f,g)}{\|g\|_2^2}$$

PDF 66, math fragment 380

$$n$$

PDF 66, math fragment 381

$$f(x),g(x)\in C[a,b]$$

PDF 66, math fragment 382

$$f$$

PDF 66, math fragment 383

$$g$$

PDF 66, math fragment 384

$$[a,b]$$

PDF 66, math fragment 385

$$\rho(x)$$

PDF 66, math fragment 386

$$\varphi_0(x),\varphi_1(x),\cdots,\varphi_n(x),\cdots$$

PDF 67, math fragment 387

$$1,\cos x,\sin x,\cos 2x,\sin 2x,\cdots$$

PDF 67, math fragment 388

$$(1,1)=2\pi,$$

PDF 67, math fragment 389

$$(\sin nx, \sin mx) = \int_{-\pi}^{\pi} \sin nx \sin mx dx = \begin{cases} \pi, & m = n, \\ 0, & m \neq n \end{cases} (n, m = 1, 2, \dots),$$

PDF 67, math fragment 390

$$(\cos nx, \cos mx) = \int_{-\pi}^{\pi} \cos nx \cos mx dx = \begin{cases} \pi, & m = n, \\ 0, & m \neq n \end{cases} (n, m = 1, 2, \dots),$$

PDF 67, math fragment 391

$$(\cos nx, \sin mx) = \int_{-\pi}^{\pi} \cos nx \sin mx dx = 0 (n, m = 1, 2, \dots).$$

PDF 67, math fragment 392

$$a_0\varphi_0(x) + a_1\varphi_1(x) + \cdots + a_{n-1}\varphi_{n-1}(x) = 0$$

PDF 67, math fragment 393

$$S(x) = a_0\varphi_0(x) + a_1\varphi_1(x) + \cdots + a_{n-1}\varphi_{n-1}(x)$$

PDF 67, math fragment 394

$$\Phi = \text{span}\{\varphi_0, \varphi_1, \dots, \varphi_{n-1}\}. \tag{3.3.9}$$

PDF 67, math fragment 395

$$G_{n-1} = G(\varphi_0, \varphi_1, \dots, \varphi_{n-1}) = \left| \begin{array}{cccc} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_1) & \cdots & (\varphi_0, \varphi_{n-1}) \\ (\varphi_1, \varphi_0) & (\varphi_1, \varphi_1) & \cdots & (\varphi_1, \varphi_{n-1}) \\ \vdots & \vdots & & \vdots \\ (\varphi_{n-1}, \varphi_0) & (\varphi_{n-1}, \varphi_1) & \cdots & (\varphi_{n-1}, \varphi_{n-1}) \end{array} \right|. \tag{3.3.10}$$

PDF 67, math fragment 396

$$\{\varphi_k\}$$

PDF 67, math fragment 397

$$[a, b]$$

PDF 67, math fragment 398

$$\rho(x)$$

PDF 67, math fragment 399

$$A_k \equiv 1$$

PDF 67, math fragment 400

$$\{\varphi_k\}$$

PDF 67, math fragment 401

$$[-\pi, \pi]$$

PDF 67, math fragment 402

$$\rho(x) \equiv 1$$

PDF 67, math fragment 403

$$\mathbf{R}^n$$

PDF 67, math fragment 404

$$f(x) \in C[a, b]$$

PDF 67, math fragment 405

$$\varphi_0(x), \varphi_1(x), \dots, \varphi_{n-1}(x)$$

PDF 67, math fragment 406

PDF 67, math fragment 407

PDF 67, math fragment 408

PDF 67, math fragment 409

PDF 67, math fragment 410

PDF 67, math fragment 411

PDF 67, math fragment 412

PDF 67, math fragment 413

PDF 67, math fragment 414

PDF 67, math fragment 415

PDF 67, math fragment 416

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PDF 67, math fragment 418

PDF 67, math fragment 419

PDF 67, math fragment 420

PDF 67, math fragment 421

PDF 67, math fragment 422

PDF 67, math fragment 423

PDF 67, math fragment 424

PDF 67, math fragment 425

$$[a,b]$$

$$a_0=a_1=\cdots=a_{n-1}=0$$

$$\varphi_0,\varphi_1,\cdots,\varphi_{n-1}$$

$$[a,b]$$

$$\{\varphi_k\}(k=0,1,\cdots)$$

$$\varphi_k$$

$$\{\varphi_k\}$$

$$1,x,x^2,\cdots,x^n,\cdots$$

$$[a,b]$$

$$\varphi_0(x),\varphi_1(x),\cdots,\varphi_{n-1}(x)$$

$$[a,b]$$

$$a_0,a_1,\cdots,a_{n-1}$$

$$C[a,b]$$

$$\{\varphi_k\}(k=0,1,\cdots,n-1)$$

$$\varphi_0(x),\varphi_1(x),\cdots,\varphi_{n-1}(x)$$

$$[a,b]$$

$$G_{n-1}\neq 0$$

$$[a,b]$$

$$f(x)\in C[a,b]$$

$$C[a,b]$$

PDF 68, math fragment 426

$$\|f - S^*\|_2^2 = \inf_{S \in \varphi} \|f - S\|_2^2 = \inf_{S \in \varphi} \int_a^b \rho(x) [f(x) - S(x)]^2 dx, \quad (3.3.11)$$

PDF 68, math fragment 427

$$I(a_0, a_1, \dots, a_n) = \int_a^b \rho(x) \left[\sum_{j=0}^n a_j \varphi_j(x) - f(x) \right]^2 dx \quad (3.3.12)$$

PDF 68, math fragment 428

$$\frac{\partial I}{\partial a_k} = 0 \quad (k = 0, 1, \dots, n),$$

PDF 68, math fragment 429

$$\frac{\partial I}{\partial a_k} = 2 \int_a^b \rho(x) \left[\sum_{j=0}^n a_j \varphi_j(x) - f(x) \right] \varphi_k(x) dx = 0 \quad (k = 0, 1, \dots, n),$$

PDF 68, math fragment 430

$$\sum_{j=0}^n (\varphi_k, \varphi_j) a_j = (f, \varphi_k) \quad (k = 0, 1, \dots, n). \quad (3.3.13)$$

PDF 68, math fragment 431

$$S^*(x) = a_0^* \varphi_0(x) + \dots + a_n^* \varphi_n(x).$$

PDF 68, math fragment 432

$$\int_a^b \rho(x) [f(x) - S^*(x)]^2 dx \leq \int_a^b \rho(x) [f(x) - S(x)]^2 dx. \quad (3.3.14)$$

PDF 68, math fragment 433

$$D = \int_a^b \rho(x) [f(x) - S(x)]^2 dx - \int_a^b \rho(x) [f(x) - S^*(x)]^2 dx$$

PDF 68, math fragment 434

$$= \int_a^b \rho(x) [S(x) - S^*(x)]^2 dx + 2 \int_a^b \rho(x) [S^*(x) - S(x)] [f(x) - S^*(x)] dx.$$

PDF 68, math fragment 435

$$\int_a^b \rho(x) [f(x) - S^*(x)] \varphi_k(x) dx = 0 \quad (k = 0, 1, \dots, n),$$

PDF 68, math fragment 436

$$D = \int_a^b \rho(x) [S(x) - S^*(x)]^2 dx \geq 0,$$

PDF 68, math fragment 437

$$\|\delta\|_2^2 = (f - S^*, f - S^*) = (f, f) - (S^*, f) = \|f\|_2^2 - \sum_{k=0}^n a_k^* (\varphi_k, f). \quad (3.3.15)$$

PDF 68, math fragment 438

$$\Phi = \text{span}\{\varphi_0, \varphi_1, \dots, \varphi_n\}$$

PDF 68, math fragment 439

$$S^*(x) \in \Phi$$

PDF 68, math fragment 440

$$S^*(x)$$

PDF 68, math fragment 441

$$f(x)$$

PDF 68, math fragment 442

$$\Phi \subseteq C[a, b]$$

PDF 68, math fragment 443

$$S^*(x)$$

PDF 68, math fragment 444

PDF 68, math fragment 445

PDF 68, math fragment 446

PDF 68, math fragment 447

PDF 68, math fragment 448

PDF 68, math fragment 449

PDF 68, math fragment 450

PDF 68, math fragment 451

PDF 68, math fragment 452

PDF 68, math fragment 453

PDF 68, math fragment 454

PDF 68, math fragment 455

PDF 68, math fragment 456

PDF 68, math fragment 457

PDF 68, math fragment 458

PDF 68, math fragment 459

PDF 68, math fragment 460

PDF 69, math fragment 461

PDF 69, math fragment 462

$$I(a_0,a_1,\ldots,a_n)$$

$$a_0,a_1,\ldots,a_n$$

$$a_0,a_1,\ldots,a_n$$

$$\varphi_0,\varphi_1,\ldots,\varphi_n$$

$$G(\varphi_0,\varphi_1,\ldots,\varphi_n)\neq 0$$

$$a_k=a_k^*(k=0,1,\ldots,n)$$

$$S^*(x)$$

$$S(x)\in\Phi$$

$$S^*(x)$$

$$a_k^*$$

$$S^*(x)$$

$$f(x)$$

$$\Phi$$

$$\delta=f(x)-S^*(x)$$

$$S\in\varphi$$

$$\Phi$$

$$\Phi$$

$$S^*(x)=a_0^*+a_1^*x+\cdots+a_n^*x^n,$$

$$(\varphi_j,\varphi_k)=\int_0^1x^{k+j}\mathrm{d}x=\frac{1}{k+j+1},$$

PDF 69, math fragment 463

$$(f,\varphi_k)=\int_0^1f(x)x^k\mathrm{d}x\equiv d_k.$$

PDF 69, math fragment 464

$$\boldsymbol{H}=\begin{pmatrix}1&1/2&\cdots&1/(n+1)\\1/2&1/3&\cdots&1/(n+2)\\\vdots&\vdots&&\vdots\\1/(n+1)&1/(n+2)&\cdots&1/(2n+1)\end{pmatrix},\tag{3.3.16}$$

PDF 69, math fragment 465

$$\boldsymbol{a}=(a_0,a_1,\cdots,a_n)^{\mathrm{T}},\quad \boldsymbol{d}=(d_0,d_1,\cdots,d_n)^{\mathrm{T}},$$

PDF 69, math fragment 466

$$d_k=(f,x^k)(k=0,1,\cdots,n),\tag{3.3.17}$$

PDF 69, math fragment 467

$$\boldsymbol{H}\boldsymbol{a}=\boldsymbol{d}$$

PDF 69, math fragment 468

$$d_0=\int_0^1\sqrt{1+x^2}\mathrm{d}x=\frac{1}{2}\ln(1+\sqrt{2})+\frac{\sqrt{2}}{2}\approx1.147,$$

PDF 69, math fragment 469

$$d_1=\int_0^1x\sqrt{1+x^2}\mathrm{d}x=\frac{1}{3}(1+x^2)^{\frac{3}{2}}\bigg|_0^1=\frac{2\sqrt{2}-1}{3}\approx0.609,$$

PDF 69, math fragment 470

$$\begin{bmatrix}1&\frac{1}{2}\\\frac{1}{2}&\frac{1}{3}\end{bmatrix}\begin{pmatrix}a_0\\a_1\end{pmatrix}=\begin{pmatrix}1.147\\0.609\end{pmatrix},$$

PDF 69, math fragment 471

$$a_0=0.934,\quad a_1=0.426,\quad S_1^*(x)=0.934+0.426x.$$

PDF 69, math fragment 472

$$\|\delta\|_2^2=(f,f)-(S_1^*,f)$$

PDF 69, math fragment 473

$$=\int_0^1(1+x^2)\mathrm{d}x-0.426d_1-0.934d_0=0.0026,$$

PDF 69, math fragment 474

$$\|\delta\|_\infty=\max_{0\leqslant x\leqslant 1}|\sqrt{1+x^2}-S_1^*(x)|\approx0.066.$$

PDF 69, math fragment 475

$$\varphi_k(x)=x^k,\rho(x)\equiv1,f(x)\in C[0,1]$$

PDF 69, math fragment 476

$$H_n$$

PDF 69, math fragment 477

$$n$$

PDF 69, math fragment 478

$$\boldsymbol{H}$$

PDF 69, math fragment 479

$$G_n=G(1,x,x^2,\cdots,x^n)$$

PDF 69, math fragment 480

$$\boldsymbol{H}$$

PDF 69, math fragment 481

$$a_k=a_k^*$$

PDF 69, math fragment 482

$$k=0,1,\cdots,n$$

PDF 69, math fragment 483

$$f(x)=\sqrt{1+x^2}$$

PDF 69, math fragment 484

$$[0,1]$$

PDF 69, math fragment 485

$$\{1,x,x^2,\cdots,x^n\}$$

PDF 69, math fragment 486

$$n$$

PDF 69, math fragment 487

$$0.0026$$

PDF 69, math fragment 488

$$d_0,d_1$$

PDF 69, math fragment 489

$$1.147,0.609$$

PDF 69, math fragment 490

$$S_1^*(x)=0.934+0.426x$$

PDF 69, math fragment 491

$$\int_0^1[\sqrt{1+x^2}-S_1^*(x)]^2\,\mathrm{d}x\approx 0.0007136324$$

PDF 69, math fragment 492

$$0.0026$$

PDF 70, math fragment 493

$$(g_j,g_k)=\int_a^b\rho(x)g_j(x)g_k(x)\mathrm{d}x=\begin{cases}0,&j\neq k,\\A_k>0,&j=k\end{cases}\quad(j,k=0,1,\cdots),$$

PDF 70, math fragment 494

$$g_0(x)=1,\quad g_n(x)=x^n-\sum_{k=0}^{n-1}\frac{(x^n,g_k)}{(g_k,g_k)}\cdot g_k(x),\quad k=1,2,\cdots.$$

PDF 70, math fragment 495

$$g_{n+1}(x)=(x-\alpha_n)g_n(x)-\beta_ng_{n-1}(x),\quad n=0,1,\cdots,$$

PDF 70, math fragment 496

$$g_0(x)=1,\quad g_{n-1}(x)=0,$$

PDF 70, math fragment 497

$$\alpha_n=\frac{(xg_n,g_n)}{(g_n,g_n)},\quad n=0,1,\cdots,$$

PDF 70, math fragment 498

$$\beta_n=\frac{(g_n,g_n)}{(g_{n-1},g_{n-1})},\quad n=1,2,\cdots.$$

PDF 70, math fragment 499

PDF 70, math fragment 500

PDF 70, math fragment 501

PDF 70, math fragment 502

PDF 70, math fragment 503

PDF 70, math fragment 504

PDF 70, math fragment 505

PDF 70, math fragment 506

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PDF 70, math fragment 510

PDF 70, math fragment 511

PDF 70, math fragment 512

PDF 70, math fragment 513

PDF 70, math fragment 514

PDF 70, math fragment 515

PDF 70, math fragment 516

PDF 70, math fragment 517

PDF 70, math fragment 518

$$g_n(x)$$

$$a_n \neq 0$$

$$n$$

$$g_0(x)$$

$$g_1(x)$$

$$\cdots$$

$$g_0(x), g_1(x), \cdots$$

$$[a, b]$$

$$\rho(x)$$

$$g_n(x)$$

$$[a, b]$$

$$\rho(x)$$

$$n$$

$$\rho(x)$$

$$[a, b]$$

$$\{1, x, x^2, \cdots, x^n, \cdots\}$$

$$g_n(x)$$

$$n$$

$$n$$

$$P_n(x) \in H_n$$

PDF 70, math fragment 519

PDF 70, math fragment 520

PDF 70, math fragment 521

PDF 70, math fragment 522

PDF 70, math fragment 523

PDF 70, math fragment 524

PDF 70, math fragment 525

PDF 70, math fragment 526

PDF 70, math fragment 527

PDF 70, math fragment 528

PDF 70, math fragment 529

PDF 70, math fragment 530

PDF 70, math fragment 531

PDF 70, math fragment 532

PDF 70, math fragment 533

PDF 70, math fragment 534

PDF 70, math fragment 535

PDF 70, math fragment 536

PDF 70, math fragment 537

PDF 70, math fragment 538

$$g_0(x), g_1(x), \cdots, g_n(x)$$

$$n \neq m$$

$$(g_n, g_m) = 0$$

$$g_n(x)$$

$$n$$

$$(xg_n, g_n) = \int_a^b xg_n^2(x)\rho(x)\mathrm{d}x.$$

$$g_0(x), g_1(x), \cdots$$

$$[a, b]$$

$$\rho(x)$$

$$g_n(x)$$

$$n \geq 1$$

$$n$$

$$(a,b)$$

$$[-1,1]$$

$$\rho(x) \equiv 1$$

$$\{1,x,x^2,\cdots,x^n,\cdots\}$$

$$k=1,2,\cdots$$

$$k$$

$$n=1,2,\cdots$$

$$g_{n-1}(x)=0$$

PDF 70, math fragment 539

$$n$$

PDF 70, math fragment 540

$$g_0(x)=1$$

PDF 70, math fragment 541

$$g_{-1}(x)=0$$

PDF 71, math fragment 542

$$P_0(x)=1,\quad P_n(x)=\frac{1}{2^nn!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\{(x^2-1)^n\}\quad (n=1,2,\cdots).$$
(3.4.1)

PDF 71, math fragment 543

$$P_n(x)=\frac{1}{2^{2n}n!}(2n)(2n-1)\cdots(n+1)x^n+a_{n-1}x^{n-1}+\cdots+a_0,$$

PDF 71, math fragment 544

$$\tilde{P}_n(x)=\frac{n!}{(2n)!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}[(x^2-1)^n].$$
(3.4.2)

PDF 71, math fragment 545

$$\int_{-1}^1P_n(x)P_m(x)\mathrm{d}x=\begin{cases}0,&m\neq n,\\ \frac{2}{2n+1},&m=n.\end{cases}$$
(3.4.3)

PDF 71, math fragment 546

$$\varphi^{(k)}(\pm1)=0\quad (k=0,1,\cdots,n-1).$$

PDF 71, math fragment 547

$$\int_{-1}^1P_n(x)Q(x)\mathrm{d}x=\frac{1}{2^nn!}\int_{-1}^1Q(x)\varphi^{(n)}(x)\mathrm{d}x=-\frac{1}{2^nn!}\int_{-1}^1Q'(x)\varphi^{(n-1)}(x)\mathrm{d}x$$

PDF 71, math fragment 548

$$=\cdots=\frac{(-1)^n}{2^nn!}\int_{-1}^1Q^{(n)}(x)\varphi(x)\mathrm{d}x.$$

PDF 71, math fragment 549

$$\int_{-1}^1P_m(x)P_n(x)\mathrm{d}x=0.$$

PDF 71, math fragment 550

$$Q(x)=P_n(x)=\frac{1}{2^nn!}\varphi^{(n)}(x)=\frac{(2n)!}{2^n(n!)^2}x^n+\cdots,$$

PDF 71, math fragment 551

$$Q^{(n)}(x)=P_n^{(n)}(x)=\frac{(2n)!}{2^nn!},$$

PDF 71, math fragment 552

$$\int_{-1}^1P_n^2(x)\mathrm{d}x=\frac{(-1)^n(2n)!}{2^{2n}(n!)^2}\int_{-1}^1(x^2-1)^n\mathrm{d}x=\frac{(2n)!}{2^{2n}(n!)^2}\int_{-1}^1(1-x^2)^n\mathrm{d}x.$$

PDF 71, math fragment 553

$$\int_0^1(1-x^2)^n\mathrm{d}x=\int_0^{\frac{\pi}{2}}\cos^{2n+1}t\mathrm{d}t=\frac{2\cdot4\cdots2n}{1\cdot3\cdots(2n+1)},$$

PDF 71, math fragment 554

$$\int_{-1}^1P_n^2(x)\mathrm{d}x=\frac{2}{2n+1}.$$

PDF 71, math fragment 555

$$P_0(x),P_1(x),\cdots,P_n(x),\cdots$$

PDF 71, math fragment 556

$$(x^2-1)^n$$

PDF 71, math fragment 557

PDF 71, math fragment 558

PDF 71, math fragment 559

PDF 71, math fragment 560

PDF 71, math fragment 561

PDF 71, math fragment 562

PDF 71, math fragment 563

PDF 71, math fragment 564

PDF 71, math fragment 565

PDF 71, math fragment 566

PDF 71, math fragment 567

PDF 71, math fragment 568

PDF 71, math fragment 569

PDF 71, math fragment 570

PDF 71, math fragment 571

PDF 71, math fragment 572

PDF 71, math fragment 573

PDF 71, math fragment 574

PDF 72, math fragment 575

PDF 72, math fragment 576

$$2n$$

$$n$$

$$x^n$$

$$a_n=\frac{(2n)!}{2^n(n!)^2}$$

$$\varphi(x)=(x^2-1)^n$$

$$Q(x)$$

$$[-1,1]$$

$$n$$

$$Q(x)$$

$$n$$

$$Q^{(n)}(x)\equiv 0$$

$$m\neq n$$

$$m=n$$

$$(x^2-1)^n$$

$$P_n(x)$$

$$2^2n!$$

$$2^nn!$$

$$a_n=(2n)!/[2^n(n!)^2]$$

$$P_n(-x)=(-1)^nP_n(x). \tag{3.4.4}$$

$$xP_n(x)=a_0P_0(x)+a_1P_1(x)+\cdots+a_{n+1}P_{n+1}(x),$$

PDF 72, math fragment 577

$$\int_{-1}^1 xP_n(x)P_k(x)dx = a_k \int_{-1}^1 P_k^2(x)dx.$$

PDF 72, math fragment 578

$$xP_n(x) = a_{n-1}P_{n-1}(x) + a_{n+1}P_{n+1}(x),$$

PDF 72, math fragment 579

$$a_{n-1} = \frac{2n-1}{2} \int_{-1}^1 xP_n(x)P_{n-1}(x)dx = \frac{2n-1}{2} \frac{2n}{4n^2-1} = \frac{n}{2n+1},$$

PDF 72, math fragment 580

$$a_{n+1} = \frac{2n+3}{2} \int_{-1}^1 xP_n(x)P_{n+1}(x)dx = \frac{2n+3}{2} \frac{2(n+1)}{(2n+1)(2n+3)} = \frac{n+1}{2n+1},$$

PDF 72, math fragment 581

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x) \quad (n = 1, 2, \dots). \tag{3.4.5}$$

PDF 72, math fragment 582

$$P_2(x) = (3x^2 - 1)/2,$$

PDF 72, math fragment 583

$$P_3(x) = (5x^3 - 3x)/2,$$

PDF 72, math fragment 584

$$P_4(x) = (35x^4 - 30x^2 + 3)/8,$$

PDF 72, math fragment 585

$$P_5(x) = (63x^5 - 70x^3 + 15x)/8,$$

PDF 72, math fragment 586

$$P_6(x) = (231x^6 - 315x^4 + 105x^2 - 5)/16,$$

PDF 72, math fragment 587

$$\vdots$$

PDF 72, math fragment 588

$$\varphi(x) = (x^2 - 1)^n$$

PDF 72, math fragment 589

$$n$$

PDF 72, math fragment 590

$$P_n(x)$$

PDF 72, math fragment 591

$$n$$

PDF 72, math fragment 592

$$P_n(x)$$

PDF 72, math fragment 593

$$n+1$$

PDF 72, math fragment 594

$$xP_n(x)$$

PDF 72, math fragment 595

$$P_k(x)$$

PDF 72, math fragment 596

PDF 72, math fragment 597

PDF 72, math fragment 598

PDF 72, math fragment 599

PDF 72, math fragment 600

PDF 72, math fragment 601

PDF 72, math fragment 602

PDF 72, math fragment 603

PDF 72, math fragment 604

PDF 72, math fragment 605

PDF 72, math fragment 606

PDF 72, math fragment 607

PDF 72, math fragment 608

PDF 72, math fragment 609

PDF 72, math fragment 610

PDF 72, math fragment 611

PDF 72, math fragment 612

PDF 72, math fragment 613

PDF 72, math fragment 614

PDF 72, math fragment 615

$$-1$$

$$1$$

$$k \leqslant n-2$$

$$xP_k(x)$$

$$n-1$$

$$a_k=0$$

$$k=n$$

$$xP_n^2(x)$$

$$a_n=0$$

$$P_0(x)=1$$

$$P_1(x)=x$$

$$P_0(x),P_1(x),P_2(x),P_3(x)$$

$$P_0(x)$$

$$P_1(x)$$

$$P_2(x)$$

$$P_3(x)$$

$$x$$

$$y$$

$$O$$

$$-1$$

PDF 72, math fragment 616

PDF 72, math fragment 617

PDF 72, math fragment 618

PDF 72, math fragment 619

PDF 72, math fragment 620

PDF 72, math fragment 621

PDF 72, math fragment 622

PDF 72, math fragment 623

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PDF 73, math fragment 629

PDF 73, math fragment 630

PDF 73, math fragment 631

PDF 73, math fragment 632

PDF 73, math fragment 633

PDF 73, math fragment 634

$$1$$

$$P_0(x)$$

$$P_1(x)$$

$$P_2(x)$$

$$P_3(x)$$

$$n$$

$$\tilde{P}_n(x)$$

$$[-1,1]$$

$$Q_n(x)$$

$$n$$

$$Q_n(x) = \tilde{P}_n(x) + \sum_{k=0}^{n-1} a_k \tilde{P}_k(x),$$

$$(Q_n, Q_n) = \int_{-1}^1 Q_n^2(x) dx = (\tilde{P}_n, \tilde{P}_n) + \sum_{k=0}^{n-1} a_k^2 (\tilde{P}_k, \tilde{P}_k) \geqslant (\tilde{P}_n, \tilde{P}_n)$$

$$T_n(x) = \cos(n \arccos x), \quad |x| \leqslant 1. \tag{3.4.6}$$

$$T_n(x) = \cos n\theta, \quad 0 \leqslant \theta \leqslant \pi.$$

$$T_0(x) = 1, \quad T_1(x) = x,$$

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad (n = 1, 2, \cdots). \tag{3.4.7}$$

$$\cos(n+1)\theta = 2\cos\theta\cos n\theta - \cos(n-1)\theta \quad (n \geqslant 1)$$

$$a_0 = a_1 = \cdots = a_{n-1} = 0$$

$$Q_n(x) \equiv \tilde{P}_n(x)$$

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$$P_n(x)$$

$$(-1,1)$$

$$n$$

$$\rho(x)=\frac{1}{\sqrt{1-x^2}}$$

$$[-1,1]$$

$$\{1,x,x^2,\cdots,x^n,\cdots\}$$

$$x=\cos\theta$$

$$x=\cos\theta$$

$$T_n(x)$$

$$T_0(x)$$

$$1$$

$$T_1(x)$$

$$x$$

$$T_2(x)$$

$$2x^2-1$$

$$T_3(x)$$

$$4x^3-3x$$

$$T_4(x)$$

$$8x^4-8x^2+1$$

$$T_5(x)$$

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$$16x^5 - 20x^3 + 5x$$

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$$T_6(x)$$

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$$32x^6 - 48x^4 + 18x^2 - 1$$

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$$T_7(x)$$

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$$64x^7 - 112x^5 + 56x^3 - 7x$$

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$$T_8(x)$$

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$$128x^8 - 256x^6 + 160x^4 - 32x^2 + 1$$

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$$T_0(x)$$

PDF 73, math fragment 663

$$T_1(x)$$

PDF 73, math fragment 664

$$T_2(x)$$

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$$T_3(x)$$

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$$x$$

PDF 73, math fragment 667

$$y$$

PDF 73, math fragment 668

$$O$$

PDF 73, math fragment 669

$$-1$$

PDF 73, math fragment 670

$$1$$

PDF 73, math fragment 671

$$1$$

PDF 73, math fragment 672

$$-1$$

PDF 73, math fragment 673

$$T_0(x)$$

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$$T_1(x)$$

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$$T_2(x)$$

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$$T_3(x)$$

PDF 73, math fragment 677

$$x = \pm 1$$

PDF 74, math fragment 678

$$\max_{-1 \leq x \leq 1} |\omega_n(x)| = \frac{1}{2^{n-1}} \max_{-1 \leq x \leq 1} |T_n(x)| = \frac{1}{2^{n-1}},$$

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$$\max_{-1 \leq x \leq 1} |f(x) - P_2^*(x)| = \min.$$

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$$f(x) - P_2^*(x) = \frac{1}{2}T_3(x) = 2x^3 - \frac{3}{2}x$$

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$$P_2^*(x) = f(x) - \frac{1}{2}T_3(x) = x^2 + \frac{7}{2}x - 1$$

PDF 74, math fragment 682

$$\int_{-1}^1 \frac{T_n(x)T_m(x) \, dx}{\sqrt{1-x^2}} = \begin{cases} 0, & n \neq m, \\ \frac{\pi}{2}, & n = m \neq 0, \\ \pi, & n = m = 0. \end{cases} \tag{3.4.8}$$

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$$\int_{-1}^1 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}} \, dx = \int_0^\pi \cos n\theta \cos m\theta \, d\theta = \begin{cases} 0, & n \neq m, \\ \frac{\pi}{2}, & n = m \neq 0, \\ \pi, & n = m = 0. \end{cases}$$

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$$T_n(x)$$

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$$2^{n-1}(n \geq 1)$$

PDF 74, math fragment 686

$$T_n(x)$$

PDF 74, math fragment 687

$$[-1, 1]$$

PDF 74, math fragment 688

$$n$$

PDF 74, math fragment 689

$$\omega_n(x) = \frac{1}{2^{n-1}}T_n(x)$$

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$$\frac{1}{2^{n-1}}$$

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$$\omega_n(x) = \frac{1}{2^{n-1}}T_n(x) = x^n - P_{n-1}^*(x)$$

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$$x_k = \cos \frac{k}{n}\pi (k = 0, 1, \cdots, n)$$

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$$T_n(x)$$

$$[-1,1]$$

$$x^n$$

$$H_{n-1}$$

$$P_{n-1}^*(x)$$

$$\omega_n(x)$$

$$f(x)=2x^3+x^2+2x-1$$

$$[-1,1]$$

$$P_2^*(x)$$

$$f(x)$$

$$[-1,1]$$

$$\{T_k(x)\}$$

$$[-1,1]$$

$$\rho(x)=1/\sqrt{1-x^2}$$

$$x=\cos\theta$$

$$\mathrm{d}x=-\sin\theta\mathrm{d}\theta$$

$$T_{2k}(x)$$

$$x$$

$$T_{2k+1}(x)$$

$$x$$

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$$x^n=2^{1-n}\sum_{k=0}^{[\frac{n}{2}]}\binom{n}{k}T_{n-2k}(x). \tag{3.4.9}$$

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$$U_n(x)=\frac{\sin[(n+1)\arccos x]}{\sqrt{1-x^2}}. \tag{3.4.10}$$

PDF 75, math fragment 715

$$\int_{-1}^1U_n(x)U_m(x)\sqrt{1-x^2}\mathrm{d}x=\int_0^\pi\sin(n+1)\theta\sin(m+1)\theta\mathrm{d}\theta$$

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$$=\left\{\begin{array}{ll}0,&m\neq n,\\ \frac{\pi}{2},&m=n,\end{array}\right.$$

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$$T_n(x)$$

PDF 75, math fragment 718

$$[-1,1]$$

PDF 75, math fragment 719

$$n$$

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$$x_k=\cos\frac{2k-1}{2n}\pi,k=1,\cdots,n$$

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$$x^n$$

PDF 75, math fragment 722

$$T_0,T_1,\cdots,T_n$$

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$$T_0=1/2$$

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$$n=1\sim 8$$

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$$1$$

PDF 75, math fragment 726

$$T_0$$

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$$x$$

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$$T_1$$

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$$x^2$$

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$$\frac{1}{2}(T_0+T_2)$$

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$$x^3$$

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$$\frac{1}{4}(3T_1 + T_3)$$

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$$x^4$$

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$$\frac{1}{8}(3T_0 + 4T_2 + T_4)$$

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$$x^5$$

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$$\frac{1}{16}(10T_1 + 5T_3 + T_5)$$

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$$x^6$$

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$$\frac{1}{32}(10T_0 + 15T_2 + 6T_4 + T_6)$$

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$$x^7$$

PDF 75, math fragment 740

$$\frac{1}{64}(35T_1 + 21T_3 + 7T_5 + T_7)$$

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$$x^8$$

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$$\frac{1}{128}(35T_0 + 56T_2 + 28T_4 + 8T_6 + T_8)$$

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$$[a, b]$$

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$$\rho(x)$$

PDF 75, math fragment 745

$$[-1, 1]$$

PDF 75, math fragment 746

$$\rho(x) = \sqrt{1 - x^2}$$

PDF 75, math fragment 747

$$x = \cos \theta$$

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$$T_0 = 1/2$$

PDF 75, math fragment 749

$$1 = T_0$$

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$$T_0 = 1$$